

Dolphin

Camera-based autonomous driving.

From driving data to on-vehicle experiments. We develop end-to-end driving systems and explore reasoning, simulation and efficient learning for autonomous driving.

CURRENT STATUS

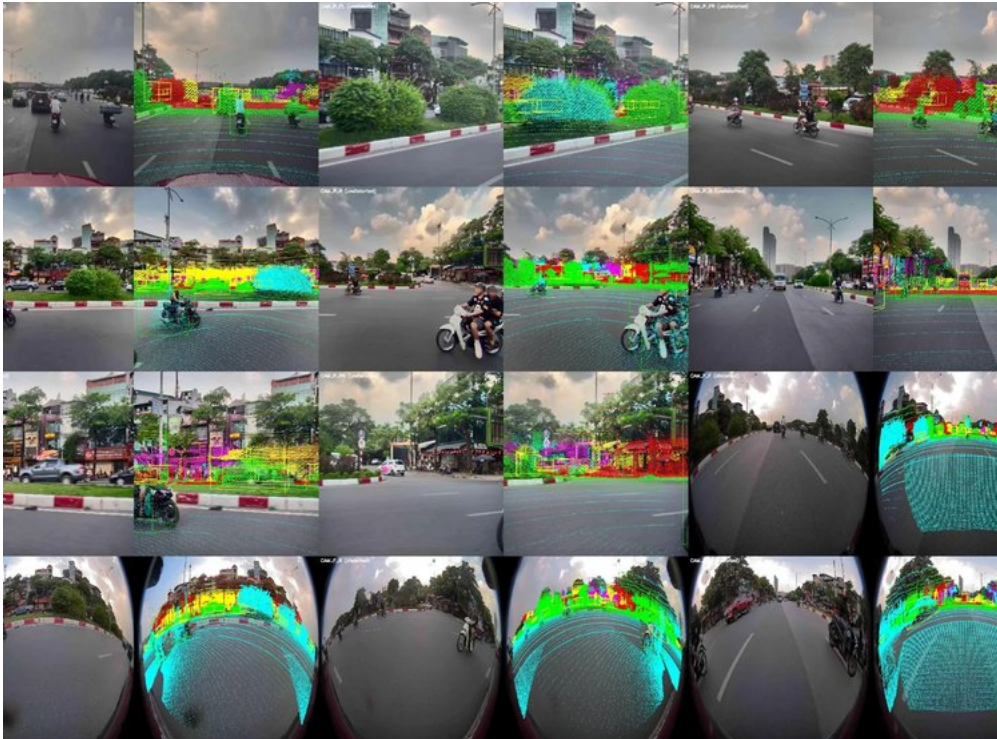
E2E: on-vehicle proof of concept.

VLA: early integration; shadow-mode evaluation planned.



Phenikaa-X research vehicle

From data to vehicle evaluation



01 Data collection & curation

Capture, calibration, timestamp alignment, semi-automatic annotation and visualization.

02 Driving model development

Perception, prediction and planning, progressing toward E2E driving and VLA guidance.

03 Real-time integration

PhenikaaX Drive Platform, efficient communication and on-vehicle execution.

04 Simulation & evaluation

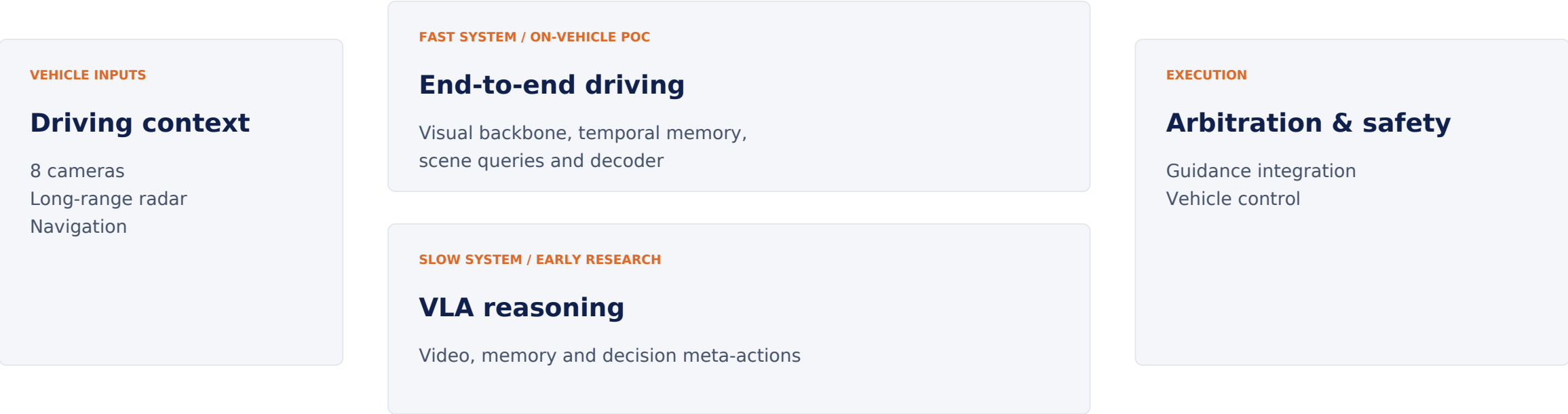
HD-map-based and camera-based simulation development, closed-loop evaluation and controlled tests.

Collect / Curate & label / Train / Simulate & test / Vehicle evaluation

SYSTEM DIRECTION

Fast E2E driving with VLA guidance

A proposed dual-system architecture for challenging driving scenarios. The reasoning module runs asynchronously alongside the E2E system.



Development direction: research to pilot in 2027.

Scientific questions with driving applications

01 Efficient end-to-end driving

Architectures and temporal representations under practical compute, latency and memory constraints.

02 Multimodal reasoning & VLA

Spatial and temporal reasoning, useful driving guidance, and decision-relevant datasets and benchmarks.

03 Data-efficient perception & learning

Data selection, limited-label learning and adaptation across traffic scenarios and operating conditions.

04 4D reconstruction & simulation

Dynamic scene reconstruction, Power Foam-based methods and Gaussian splatting for closed-loop evaluation.

05 Learning better driving decisions

Trajectory diversity and policy improvement from data, with reduced dependence on dedicated simulators.

Selected ongoing and planned directions; related research topics remain open for discussion.

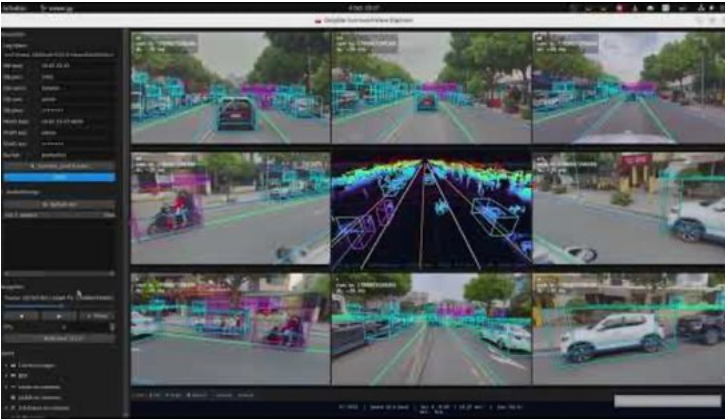
Selected project videos



End-to-end driving

On-vehicle proof of concept.

Watch video



Data visualization

Driving data and annotation workflow.

Watch video



Dolphin on VTV

A feature on the vehicle and project.

Watch video

Select a button to open the corresponding video online.

PEOPLE & COLLABORATION

Explore a research direction with us

We welcome research advisory, joint research and mentoring collaborations in autonomous driving. Our team contributes implementation, experiments and a growing platform for evaluation.

Lê Anh Sơn

Company Director

Vũ Thành Đạt

Head of AV Research / Dolphin Lead

Lý Hoàng Hiệp

Project Coordinator

Phạm Hoàng Sơn

Project Coordinator

CORE TEAMS

E2E / Perception · Reasoning / Planning · Sensing / System
Navigation · Localization · Control

Shared support: mechatronics, mechanical engineering and web / app development.

CONTACT

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Visit Phenikaa-X